

3. Thresholding

(a) Global thresholding in segmentation

Global thresholding is a segmentation technique that converts a grey-scale image into a binary image using a single threshold value T applied uniformly across the whole image:

$$g(x, y) = 1 \text{ if } f(x, y) \geq T, \quad g(x, y) = 0 \text{ if } f(x, y) < T$$

Pixels are grouped into two classes — object (foreground) and background — based purely on intensity. T is typically chosen from the histogram (e.g. the valley between two peaks, or an iterative/Otsu method). It works well when illumination is uniform, but performs poorly on images with uneven lighting.

(b) Apply threshold $T = 128$

No specific pixel values were supplied, so the sample set below is used.

Pixel value	30	60	90	120	150	180	210	240
Binary output ($T = 128$)	0	0	0	0	1	1	1	1

Pixels below 128 are assigned 0 (background); pixels at or above 128 are assigned 1 (foreground).